Quadratic Reciprocity Law

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January 2026

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1 Notation

Let p be a prime number and K be a field. We denote

$$K^* = K \setminus \{0\},$$

and

$$\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z} = \{\bar{1}, \bar{2}, \ldots, \bar{p}\}.$$

The characteristic of a field K, denoted by $\mathrm{char}(K)$, is the smallest positive integer such that $\mathrm{char}(K) \cdot 1_K = 0$.

2 Finite Field

Lemma 2.1. *If $char(K) = p$, the map $\sigma : x \mapsto x^p$ is an isomorphism from K onto K^p.*

Proof. First, observe that $\sigma(1) = 1^p = 1$ and $\sigma(0) = 0^p = 0$. It is clear that

$$\sigma(xy) = (xy)^p = x^p y^p = \sigma(x)\sigma(y)$$

and

$$\sigma(x+y) = (x+y)^p = x^p + y^p = \sigma(x) + \sigma(y).$$

Hence, σ is a ring homomorphism. Now, we will show that σ is injective. Suppose there exist $a, b \in K$ with $a \neq b$ such that $\sigma(a) = \sigma(b)$. Then we have

$$\sigma(a) - \sigma(b) = \sigma(a-b) = 0.$$

For any $c \in K$, consider that

$$\sigma(c) = \sigma((a-b)^{-1}(a-b)c) = 0.$$

In particular, if $c = 1$, then $\sigma(1) = 0$, which contradicts $\sigma(1) = 1$. Hence, we must have $a = b$, proving that σ is injective.
Since $K^p \subseteq K$, we have

$$\mathrm{Card}(K^p) \leq \mathrm{Card}(K).$$

Furthermore, since σ is injective, we also have

$$\mathrm{Card}(K) \leq \mathrm{Card}(K^p),$$

which implies $K^p = K$. Therefore, σ is an isomorphism from K onto K^p. □

Theorem 1.

(i) The characteristic of a finite field K is a prime number $p > 0$; if $f = [K : \mathbb{F}_p]$, then the number of elements in K is $q = p^f$.

(ii) Let $q = p^f$ and let Ω be an algebraically closed field of characteristic p. There exists a unique subfield $\mathbb{F}_q$ of Ω having exactly q elements. It is the set of roots of the polynomial $X^q - X$.

(iii) All finite fields with $q = p^f$ elements are isomorphic to $\mathbb{F}_q$.

Proof. (i) We first show that $\mathrm{char}(K) > 0$. If $\mathrm{char}(K)$ were 0, the elements of the set

$$\{n \cdot 1_K : n \in \mathbb{N}\} \subset K$$

would all be distinct, implying that K is infinite, which contradicts the finiteness of K. Hence, $\mathrm{char}(K)$ exists as a positive integer for every finite field K.
Now, we show that $\mathrm{char}(K)$ is prime. Suppose K is a finite field with $\mathrm{char}(K) = n = ab$ for some integers a, b such that $1 < a, b < n$. Then

$$0 = n \cdot 1_K = (ab) \cdot 1_K = (a \cdot 1_K)(b \cdot 1_K),$$

so

$$(a \cdot 1_K)^{-1} \cdot 0 = 0 = b \cdot 1_K.$$

This implies $b = n$ or $b = 0$, which contradicts the assumption that $1 < b < n$. Hence, n must be prime. Since $f = [K : \mathbb{F}_p]$, it directly follows that $\mathrm{Card}(K) = p^f$.
(ii) Since $\mathrm{char}(\Omega) = p$, the previous lemma implies that $\sigma^f : x \mapsto x^q$ is an automorphism of Ω. Its fixed field

$$\mathbb{F}_q = \{x \in \Omega : x^q = x\}$$

is therefore a subfield of Ω.
We claim that if $\gcd(f'(x), f(x)) = 1$, then all roots of $f(x) = 0$ are distinct. Suppose for the sake of contradiction that they are not distinct, meaning

$$f(x) = (x - a_1)^{b_1}(x - a_2)^{b_2} \cdots (x - a_n)^{b_n}$$

for some $a_i \in \Omega$, with at least one exponent $b_i > 1$. Then

$$f'(x) = \sum_{i=1}^{n} \frac{b_i \cdot f(x)}{x - a_i}.$$

If $b_j > 1$, then $(x - a_j)$ divides both $f(x)$ and $f'(x)$, which means $(x - a_j) \mid \gcd(f'(x), f(x))$, contradicting $\gcd(f'(x), f(x)) = 1$.
Now, consider $f(x) = X^q - X$. Then

$$f'(x) = q \cdot x^{q-1} - 1 = p^f \cdot x^{q-1} - 1 = -1,$$

so $\gcd(f(x), f'(x)) = 1$. Since Ω is algebraically closed, $f(x) = 0$ therefore has exactly q distinct roots in Ω.
Suppose K is any subfield of Ω with q elements. Then $x^{q-1} = 1$ for every $x \in K \setminus \{0\}$, so $x^q = x$ for every $x \in K$. Hence $K \subseteq \mathbb{F}_q$, and since $\mathrm{Card}(K) = \mathrm{Card}(\mathbb{F}_q)$, we conclude $K = \mathbb{F}_q$. This proves both existence and uniqueness.
(iii) By part (ii), any finite field with $q = p^f$ elements coincides with $\mathbb{F}_q$ inside an algebraic closure, and hence is isomorphic to $\mathbb{F}_q$. □

3 The Multiplicative Group of a Finite Field

Lemma 3.1.

(i) If $n \in \mathbb{N}$, then $n = \sum_{d|n} \phi(d)$, where $\phi(d)$ is the number of integers x such that $1 \leq x \leq d$ and $\gcd(x, d) = 1$.

(ii) For the finite group $\mathbb{F}_p^*$, there exists $t \in \mathbb{F}_p$ such that $\{t, t^2, \ldots, t^{p-1}\} = \mathbb{F}_p^*$.

Proof. (i) Consider the set of fractions $\{1/n, 2/n, \ldots, n/n\}$, which contains exactly n elements. Reduce each fraction m/n to its lowest terms k/d, where $\gcd(k, d) = 1$. For every divisor d of n, exactly $\phi(d)$ of these fractions reduce to a fraction with denominator d. Hence,

$$n = \sum_{d|n} \phi(d).$$

(ii) If $a \in \mathbb{F}_p^*$ satisfies $a^n = a^m = 1$ for some $n, m \in \mathbb{N}$ with $n > m$, then $a^{n-m} = 1$. Iterating this process (similar to the Euclidean algorithm) yields $a^{\gcd(n,m)} = 1$.
We prove by induction on $d \mid p-1$ that the equation $x^d = 1$ has exactly $\phi(d)$ solutions of exact order d (i.e., satisfying $x^e \neq 1$ for $1 \leq e < d$).
For $d = 1$, the equation $x = 1$ clearly has exactly one such solution, and $\phi(1) = 1$.
Suppose the claim holds for every proper divisor d of m, where m is also a divisor of $p-1$. Among the m roots of $x^m = 1$, the number of roots of exact order d, for each proper divisor $d < m$ of m, is exactly $\phi(d)$ by the inductive hypothesis. By part (i),

$$m = \sum_{d|m} \phi(d),$$

so the remaining

$$m - \sum_{\substack{d|m \\ d<m}} \phi(d) = \phi(m)$$

roots have exact order m. This completes the induction; taking $m = p-1$ proves part (ii). □

Theorem 2. *The multiplicative group $\mathbb{F}_q^*$ of a finite field $\mathbb{F}_q$ is cyclic of order $q-1$.*

Proof. Applying part (ii) of the previous lemma with $p-1$ replaced by $q-1$, there exists an element of $\mathbb{F}_q^*$ of exact order $q-1$. Hence, $\mathbb{F}_q^*$ is cyclic of order $q-1$. □

4 Power Sums

Lemma 4.1. *Let $u \geq 0$ be an integer, and let*

$$S(X^u) = \sum_{x \in K} x^u,$$

where K is a finite field with $q = p^f$ elements. Then

$$S(X^u) = \begin{cases} -1, & \text{if } u \geq 1 \text{ and } (q-1) \mid u, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. If $u = 0$, then $x^u = 1$ for every $x \in K$, so $S(X^0) = q = 0$ in K.
If $u \geq 1$ and $(q-1) \mid u$, then $x^u = 1$ for every $x \in K \setminus \{0\}$, and $x^u = 0$ for $x = 0$. Hence,

$$S(X^u) = (q-1) \cdot 1 = -1.$$

If $u \geq 1$ and $(q-1) \nmid u$, choose a generator t of $K \setminus \{0\}$ such that $t^u \neq 1$. The map $g : K \setminus \{0\} \to K \setminus \{0\}$ given by $g(x) = tx$ is a bijection, since $g(x_1) = g(x_2)$ implies $x_1 = x_2$, and every $z \in K \setminus \{0\}$ is the image of $t^{-1}z \in K \setminus \{0\}$. Hence,

$$t^u S(X^u) = \sum_{x \in K \setminus \{0\}} (tx)^u = \sum_{x \in K \setminus \{0\}} x^u = S(X^u),$$

which implies $(1 - t^u)S(X^u) = 0$. Since $t^u \neq 1$, we conclude that $S(X^u) = 0$. □

5 The Chevalley–Warning Theorem

Theorem 3 (Chevalley–Warning)**.** *Let $\{f_\alpha\}_{\alpha\in A} \subset K[X_1,\dots,X_n]$ be a family of polynomials in n variables such that*

$$\sum_{\alpha\in A} \deg(f_\alpha) < n,$$

and let V be the set of their common zeros in K^n. Then

$$\mathit{Card}(V) \equiv 0 \pmod{p}.$$

Proof. Let

$$P(x) = \prod_{\alpha\in A} \bigl(1 - f_\alpha(x)^{q-1}\bigr).$$

If $x \in V$, then $f_\alpha(x) = 0$ for every α, which implies $P(x) = 1$. If $x \notin V$, then $f_\alpha(x) \neq 0$ for some α; thus $f_\alpha(x)^{q-1} = 1$ and $P(x) = 0$. Hence, P is the characteristic function of V.
For a polynomial f, write $S(f) = \sum_{x\in K^n} f(x)$. Then

$$S(P) \equiv \mathrm{Card}(V) \pmod{p}.$$

Since $\sum_{\alpha\in A} \deg(f_\alpha) < n$, we have $\deg(P) < n(q-1)$, so $P(x)$ is a linear combination of monomials

$$X^u = X_1^{u_1} X_2^{u_2} \cdots X_n^{u_n}, \qquad \sum_{i=1}^{n} u_i < n(q-1).$$

For each such monomial,

$$S(X^u) = \prod_{k=1}^{n} \Bigl(\sum_{X_k\in K} X_k^{u_k} \Bigr).$$

By the pigeonhole principle, there exists some j such that $u_j < q-1$. Hence, by the lemma on power sums, $\sum_{X_j\in K} X_j^{u_j} = 0$. Thus, $S(X^u) = 0$ for every such monomial, meaning $S(P) = 0$, and therefore

$$\mathrm{Card}(V) \equiv S(P) \equiv 0 \pmod{p}.$$

□

Corollary. *If $\sum_{\alpha\in A} \deg(f_\alpha) < n$ and each f_α has no constant term, then the polynomials f_α have a non-trivial common zero.*

Corollary. *Every quadratic form in at least 3 variables over K has a non-trivial zero.*

6 Quadratic Reciprocity Law

Definition 6.1 (Legendre symbol)**.** *Let $p > 2$ be an odd prime. For $a \in \mathbb{F}_p$, define*

$$\left(\frac{a}{p}\right) = \begin{cases} 1, & \text{if } a \in (\mathbb{F}_p^*)^2, \\ -1, & \text{if } a \notin (\mathbb{F}_p^*)^2, \\ 0, & \text{if } a = 0. \end{cases}$$

Lemma 6.1. *Let $a \in \mathbb{F}_p^*$ and let $y \in \overline{\mathbb{F}_p}$ satisfy $y^2 = a$. Then*

$$\left(\frac{a}{p}\right) = y^{p-1} = a^{\frac{p-1}{2}} \in \mathbb{F}_p.$$

Proof. Since $(y^{p-1})^2 = a^{p-1} = 1$, we have $y^{p-1} = \pm 1$. Moreover,

$$\left(\frac{a}{p}\right) = 1 \iff y \in \mathbb{F}_p \iff y^p = y \iff y^{p-1} = 1.$$

Hence, $\left(\frac{a}{p}\right) = y^{p-1} = a^{\frac{p-1}{2}}$. □

Corollary. *For $a, b \in \mathbb{F}_p$,*

$$\left(\frac{a}{p}\right)\left(\frac{b}{p}\right) = a^{\frac{p-1}{2}} b^{\frac{p-1}{2}} = (ab)^{\frac{p-1}{2}} = \left(\frac{ab}{p}\right).$$

Theorem 4. *The following formulas hold:*

$$\left(\frac{1}{p}\right) = 1, \qquad \left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}, \qquad \left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}.$$

Proof. By the lemma, $\left(\frac{1}{p}\right) = 1^{\frac{p-1}{2}} = 1$ and $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$.

For $\left(\frac{2}{p}\right)$, let α be a primitive 8th root of unity in an algebraic closure Ω of $\mathbb{F}_p$, and set $y = \alpha + \alpha^{-1}$. Then

$$y^2 = 2 + \alpha^2 + \alpha^{-2} = 2 + \alpha^2(1 + \alpha^4) = 2,$$

so by the lemma,

$$\left(\frac{2}{p}\right) = y^{p-1} = 2^{\frac{p-1}{2}}.$$

Now,

$$y^p = (\alpha + \alpha^{-1})^p = \alpha^p + \alpha^{-p}.$$

If $p \equiv \pm 1 \pmod 8$, then $y^p = \alpha + \alpha^{-1} = y$, so $y^{p-1} = 1$ and $\left(\frac{2}{p}\right) = 1$.

If $p \equiv \pm 3 \pmod 8$, then

$$y^p = \alpha^3 + \alpha^{-3} = \alpha^{-5} + \alpha^5 = -\alpha^{-1} - \alpha = -y,$$

so $y^{p-1} = -1$ and $\left(\frac{2}{p}\right) = -1$. In all four cases, $\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}$. □

Theorem 5 (Quadratic Reciprocity Law)**.** *For any two distinct odd primes p and l,*

$$\left(\frac{l}{p}\right) = \left(\frac{p}{l}\right)(-1)^{\frac{l-1}{2}\cdot\frac{p-1}{2}}.$$

To prove this theorem, we introduce the *Gauss sum*. Let Ω be an algebraic closure of $\mathbb{F}_p$, and let $w \in \Omega$ be a primitive l-th root of unity. For $x \in \mathbb{F}_l$, the element w^x is well-defined since $w^l = 1$. Define

$$y = \sum_{x \in \mathbb{F}_l} \left(\frac{x}{l}\right) w^x.$$

Lemma 6.2.

(i) $y^2 = (-1)^{\frac{l-1}{2}} l$.

(ii) $y^{p-1} = \left(\frac{p}{l}\right)$.

Proof. (i) By direct calculation, we have:

$$\begin{aligned}
y^2 = \Big(\sum_{x \in \mathbb{F}_l} \left(\frac{x}{l}\right) w^x\Big)^2 &= \sum_{x=1}^{l-1} \sum_{z=0}^{l-1} \left(\frac{xz}{l}\right) w^{x+z} \\
&= \sum_{x=1}^{l-1} \sum_{t=0}^{l-1} \left(\frac{x(t-x)}{l}\right) w^t \qquad (\text{set } x+z=t) \\
&= \sum_{x=1}^{l-1} \sum_{t=0}^{l-1} \left(\frac{-x^2}{l}\right)\left(\frac{1-x^{-1}t}{l}\right) w^t \\
&= (-1)^{\frac{l-1}{2}} \sum_{x=1}^{l-1} \sum_{t=0}^{l-1} \left(\frac{1-x^{-1}t}{l}\right) w^t \\
&= (-1)^{\frac{l-1}{2}} \sum_{t=0}^{l-1} w^t \Big(\sum_{x=1}^{l-1} \left(\frac{1-x^{-1}t}{l}\right)\Big) \\
&= (-1)^{\frac{l-1}{2}} \Big(\sum_{x=1}^{l-1} \left(\frac{1}{l}\right) + \sum_{t=1}^{l-1} w^t \Big(\sum_{x=1}^{l-1} \left(\frac{1-x^{-1}t}{l}\right)\Big)\Big) \qquad (\text{set } s = 1-x^{-1}t) \\
&= (-1)^{\frac{l-1}{2}} \Big(l-1 + \sum_{t=1}^{l-1} w^t \Big(\sum_{s=0}^{l-1} \left(\frac{s}{l}\right) - \left(\frac{1}{l}\right)\Big)\Big) \\
&= (-1)^{\frac{l-1}{2}} \Big(l-1 - \left(\frac{1}{l}\right) \sum_{t=1}^{l-1} w^t\Big) \\
&= (-1)^{\frac{l-1}{2}} \Big(l-1 - \sum_{t=1}^{l-1} w^t\Big) = (-1)^{\frac{l-1}{2}} \cdot l.
\end{aligned}$$

Hence, $y^2 = (-1)^{\frac{l-1}{2}} \cdot l$.
(ii)

$$
\begin{aligned}
y^p = \Big(\sum_{x=0}^{l-1}\left(\frac{x}{l}\right)w^x\Big)^p &= \sum_{x=1}^{l-1}\left(\frac{x}{l}\right)^p w^{xp} \\
&= \sum_{x=1}^{l-1}\left(\frac{x}{l}\right)w^{xp} \qquad (\text{set } xp = z) \\
&= \sum_{z=1}^{l-1}\left(\frac{zp^{-1}}{l}\right)w^z \\
&= \left(\frac{p^{-1}}{l}\right)\sum_{z=1}^{l-1}\left(\frac{z}{l}\right)w^z \\
&= \left(\frac{p}{l}\right)\cdot y.
\end{aligned}
$$

Hence, $y^{p-1} = \left(\frac{p}{l}\right)$.
Combining the two parts, we obtain:

$$(y^2)^{\frac{p-1}{2}} = (-1)^{\frac{l-1}{2}\cdot\frac{p-1}{2}} l^{\frac{p-1}{2}} = y^{p-1} = \left(\frac{p}{l}\right),$$

so

$$(-1)^{\frac{l-1}{2}\cdot\frac{p-1}{2}}\left(\frac{l}{p}\right) = \left(\frac{p}{l}\right),$$

which is the Quadratic Reciprocity Law. □

Example 6.1.

$$
\begin{aligned}
\left(\frac{67}{97}\right) = \left(\frac{97}{67}\right) = \left(\frac{30}{67}\right) &= \left(\frac{2}{67}\right)\left(\frac{3}{67}\right)\left(\frac{5}{67}\right) \\
&= -\left(\frac{3}{67}\right)\left(\frac{5}{67}\right) \\
&= \left(\frac{67}{3}\right)\left(\frac{67}{5}\right) \\
&= \left(\frac{1}{3}\right)\left(\frac{2}{5}\right) = \left(\frac{2}{5}\right) = -1.
\end{aligned}
$$

7 Hilbert Symbol

Definition 7.1. *Let p be a prime number or ∞. Let Z_p denote the ring of p-adic integers and Q_p denote its fraction field, with the convention that $Q_\infty = \mathbb{R}$.*

$$Z_p = \Big\{(a_n) \in \prod_{n\in\mathbb{Z}^+} \mathbb{Z}/p^n\mathbb{Z} \ : \ a_{n+1} \equiv a_n \pmod{p^n} \ \ \forall\ n \in \mathbb{Z}^+\Big\}.$$

$$Q_p = \left\{ \frac{a}{b} : a, b \in Z_p,\ b \neq 0 \right\}.$$

Let $\nu_p(a)$ be the unique integer such that $a = p^{\nu_p(a)} \frac{u}{v}$ with $p \nmid u$ and $p \nmid v$; we set $\nu_p(0) = +\infty$. Define a valuation on Q_p by

$$|x|_p = \begin{cases} p^{-\nu_p(x)} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

For example, $\left|\frac{9}{8}\right|_2 = 2^3 = 8$, $\left|\frac{8}{9}\right|_2 = 2^{-3} = \frac{1}{8}$, and $\left|\frac{8}{9}\right|_7 = 1$. In particular, $Q_p^* = Q_p \setminus \{0\}$, and the units of Z_p are given by

$$Z_p^* = \{x \in Q_p : |x|_p = 1\}.$$

The Hilbert symbol $(\,\cdot\,,\,\cdot\,)_p : Q_p^* \times Q_p^* \to \{\pm 1\}$ is defined by

$$(a,b)_p = \begin{cases} 1 & \text{if there exists a non-zero } (z,x,y) \in Q_p^3 \text{ with } z^2 = ax^2 + by^2, \\ -1 & \text{otherwise.} \end{cases}$$

Here, we record some basic properties of the Hilbert symbol.

Proposition 7.1. *Let $a \in Q_p^*$. Then $(a,b)_p = 1$ if and only if $a \in NQ_p(\sqrt{b}) = \{x^2 - by^2 : (x,y) \in Q_p \times Q_p\}$.*

Proof. Suppose $(a,b)_p = 1$.
Then there exists a non-zero solution (z_0, x_0, y_0) such that $z_0^2 = ax_0^2 + by_0^2$. If $x_0 \neq 0$, then

$$a = (z_0/x_0)^2 - b(y_0/x_0)^2 \in NQ_p(\sqrt{b}).$$

If $x_0 = 0$, then $b = (z_0/y_0)^2$ (note that $y_0 \neq 0$, otherwise the solution would be trivial), and one can directly verify that $x = (1+4a)/4$ and $y = y_0(1-4a)/(4z_0)$ is a solution to $x^2 - by^2 = a$.
Conversely, if $a = \alpha^2 - b\beta^2 \in NQ_p(\sqrt{b})$, then

$$z^2 = ax^2 + by^2$$

becomes

$$z^2 = (\alpha^2 - b\beta^2)x^2 + by^2,$$

which has the non-trivial solution $(\alpha, 1, \beta)$. Hence, $(a,b)_p = 1$. □

Proposition 7.2. *Let $a, a', b, c \in Q_p^*$. Then:*

(i) $(a,b)_p = (b,a)_p$ *and* $(a,c^2)_p = 1$.

(ii) $(a,-a)_p = 1$ *and* $(a,1-a)_p = 1$.

(iii) $(a,b)_p = 1 \Rightarrow (a'a,b)_p = (a',b)_p$.

(iv) $(a,b)_p = (a,-ab)_p = (a,(1-a)b)_p$.

Proof. (i) If $(a,b)_p = 1$ with solution (z_0, x_0, y_0), then (z_0, y_0, x_0) solves the equation for $(b,a)_p$, so $(b,a)_p = 1$. If instead $(a,b)_p = -1$ and $(b,a)_p = 1$, this contradicts the symmetry established by the previous observation. Hence, $(a,b)_p = (b,a)_p$ in general.
For $(a, c^2)_p$, the non-trivial triple $(c, 0, 1)$ solves

$$z^2 = ax^2 + c^2y^2,$$

so $(a, c^2)_p = 1$.
(ii) The triple $(0, 1, 1)$ solves

$$z^2 = ax^2 - ay^2,$$

so $(a, -a)_p = 1$. The triple $(1, 1, 1)$ solves $z^2 = ax^2 + (1-a)y^2$, so $(a, 1-a)_p = 1$.
(iii) By the previous proposition, $(a,b)_p = 1$ means

$$a = x_0^2 - by_0^2$$

for some $x_0, y_0 \in Q_p$. If $(a', b)_p = 1$, write $a' = x_1^2 - by_1^2$. Then

$$\begin{aligned} aa' = (x_0^2 - by_0^2)(x_1^2 - by_1^2) &= x_0^2x_1^2 - b(x_0^2y_1^2 + x_1^2y_0^2) + b^2y_0^2y_1^2 \\ &= (x_0x_1 + by_0y_1)^2 - b(x_0y_1 + x_1y_0)^2 \in NQ_p(\sqrt{b}), \end{aligned}$$

so $(a'a, b)_p = 1 = (a', b)_p$.
If instead $(a,b)_p = -1$, we argue by contradiction. Suppose $(aa', b)_p = 1$, and write $aa' = n_0^2 - bm_0^2$ and $a = n_1^2 - bm_1^2$. Then

$$\begin{aligned} a' = \frac{aa'}{a} &= \frac{(n_0^2 - bm_0^2)(n_1^2 - bm_1^2)}{(n_1^2 - bm_1^2)^2} \\ &= \frac{(n_0n_1 + bm_0m_1)^2 - b(n_0m_1 + n_1m_0)^2}{(n_1^2 - bm_1^2)^2} \\ &= \Big(\frac{n_0n_1 + bm_0m_1}{n_1^2 - bm_1^2}\Big)^2 - b\Big(\frac{n_0m_1 + n_1m_0}{n_1^2 - bm_1^2}\Big)^2 \in NQ_p(\sqrt{b}), \end{aligned}$$

so $(a', b)_p = 1$, contradicting $(a,b)_p = -1$ (by applying the first case with the roles reversed). Hence, $(aa', b)_p = (a', b)_p$ in this case as well.
(iv) By parts (i) and (ii),

$$(a, -a)_p = (-a, a)_p = 1 \Rightarrow (a, -ab)_p = (-ab, a)_p = (b, a)_p = (a, b)_p,$$

$$(a, 1-a)_p = (1-a, a)_p = 1 \Rightarrow (a, (1-a)b)_p = ((1-a)b, a)_p = (b, a)_p = (a, b)_p,$$

using part (iii) in each implication. Hence, $(a,b)_p = (a, -ab)_p = (a, (1-a)b)_p$. □

Theorem 6 (Hensel's Lemma)**.** *Let $f(x) \in Z_p[X]$ and suppose $a \in Z_p$ satisfies $f(a) \equiv 0 \pmod{p}$ and $f'(a) \not\equiv 0 \pmod{p}$. Then there exists a unique $\alpha \in Z_p$ such that $f(\alpha) = 0$ and $\alpha \equiv a \pmod{p}$.*

Proof. See Serre, Chapter 2. □

Theorem 7 (Generalized Hensel's Lemma)**.** *Let $f(x) \in Z_p[X]$ and suppose $a \in Z_p$ satisfies*

$$|f(a)|_p < |f'(a)|_p^2.$$

Then there exists a unique $\alpha \in Z_p$ such that $f(\alpha) = 0$ and $|\alpha - a|_p < |f'(a)|_p$. Moreover,

$$|\alpha - a|_p = |f(a)/f'(a)|_p < |f'(a)|_p, \qquad |f'(\alpha)|_p = |f'(a)|_p.$$

Proof. See Serre, Chapter 2. □

Theorem 8. *If $p = \infty$, then $(a, b)_p = 1$ if $a > 0$ or $b > 0$, and $(a, b)_p = -1$ if $a < 0$ and $b < 0$.*
If $p < \infty$, write $a = p^\alpha u$ and $b = p^\beta v$ with $u, v \in U = Z_p^*$. Then

$$(a, b)_p = (-1)^{\alpha\beta\cdot\epsilon(p)} \left(\frac{u}{p}\right)^\beta \left(\frac{v}{p}\right)^\alpha \qquad \text{if } p \neq 2,$$
$$(a, b)_p = (-1)^{\epsilon(u)\epsilon(v)+\alpha\omega(v)+\beta\omega(u)} \qquad \text{if } p = 2,$$

where $\epsilon(u) = \frac{u-1}{2}$ and $\omega(u) = \frac{u^2-1}{8}$.

Proof. Write $(a, b) = (p^\alpha u, p^\beta v)$.
For $p = \infty$, it is clear that $z^2 = ax^2 + by^2$ has the solution $(\sqrt{a}, 1, 0)$ or $(\sqrt{b}, 0, 1)$ when $a > 0$ or $b > 0$. If $a < 0$ and $b < 0$, then $ax^2 + by^2 \leq 0$ while $z^2 \geq 0$, so $(a, b)_p = -1$.
Now suppose $p \neq 2$. We first show that

$$u \text{ is a square in } Z_p^* \iff \left(\frac{u}{p}\right) = 1.$$

If u is a square in Z_p^*, this is immediate. Conversely, suppose $\left(\frac{u}{p}\right) = 1$, so $u \equiv a^2 \pmod{p}$ for some a. Let $f(x) = x^2 - u$. Then

$$f(a) = a^2 - u \equiv 0 \pmod{p}, \qquad f'(a) = 2a \not\equiv 0 \pmod{p}.$$

By Hensel's Lemma, there exists $\alpha \in Z_p$ with $f(\alpha) = 0$, meaning $\alpha^2 = u$; hence u is a square in Z_p^*.
To prove the formula for $p \neq 2$, it suffices, due to the parity of α and β, to check the three cases $(\alpha, \beta) = (0, 0), (0, 1), (1, 1)$.
Case $(\alpha, \beta) = (0, 0)$. The equation

$$z^2 - ux^2 - vy^2 = 0$$

has a non-zero solution in $\mathbb{F}_p$ by the Chevalley–Warning Theorem, and hence, by Hensel's Lemma, a non-zero solution in Z_p. Thus, $(u, v)_p = 1$.
Case $(\alpha, \beta) = (0, 1)$. We must show $(u, pv)_p = \left(\frac{u}{p}\right)$. Since $(u, v)_p = 1$, Proposition 7.2 gives $(u, pv)_p = (u, p)_p$.

If u is a square in Z_p, then $\left(\frac{u}{p}\right) = 1 = (u,p)_p$ as shown above.

If u is not a square in Z_p, then $\left(\frac{u}{p}\right) = -1$. The equation $z^2 - px^2 - uy^2 = 0$ has a non-zero solution in Q_p if and only if it has a solution $(z_0, x_0, y_0) \in Z_p^3$ with $z_0, y_0 \in Z_p^*$, which would force $\left(\frac{u}{p}\right) = 1$. Since this fails, $z^2 - px^2 - uy^2 = 0$ has no non-zero solution in Q_p, so $(p,u)_p = -1 = (u,p)_p = \left(\frac{u}{p}\right)$.

Case $(\alpha, \beta) = (1,1)$. We must show $(pu, pv)_p = (-1)^{\epsilon(p)}\left(\frac{u}{p}\right)\left(\frac{v}{p}\right)$. Indeed,

$$(pu, pv)_p = (pu, -p^2uv)_p = (pu, -uv)_p = \left(\frac{-uv}{p}\right) = (-1)^{\epsilon(p)}\left(\frac{u}{p}\right)\left(\frac{v}{p}\right).$$

This completes the proof for $p \neq 2$.
Now suppose $p = 2$. We first show that

$$u \text{ is a square in } Z_2^* \iff u \equiv 1 \pmod 8.$$

If u is a square, then $u \equiv 1^2 \equiv 3^2 \equiv 5^2 \equiv 7^2 \equiv 1 \pmod 8$. Conversely, if $u \equiv 1 \pmod 8$, let $f(x) = x^2 - u$. Then

$$|f(1)|_2 = |1-u|_2 \leq \tfrac{1}{8}, \qquad |f'(1)|_2 = |2|_2 = \tfrac{1}{2},$$

so $|f(1)|_2 < |f'(1)|_2^2$, and the Generalized Hensel's Lemma gives a root of $x^2 - u = 0$ in Z_2.
As before, it suffices to consider α and β modulo 2.
Case $\alpha = \beta = 0$. We must show $(u,v)_2 = 1$ if $u \equiv 1 \pmod 4$ or $v \equiv 1 \pmod 4$, and $(u,v)_2 = -1$ otherwise.
Suppose $u \equiv 1 \pmod 4$, so $u \equiv 1$ or $5 \pmod 8$. If $u \equiv 1 \pmod 8$, then u is a square in Z_2, so $(u,v)_2 = 1$. If $u \equiv 5 \pmod 8$, then $u + 4v \equiv 1 \pmod 8$, so $w^2 = u + 4v$ for some $w \in Z_2$, and $z^2 - ux^2 - vy^2 = 0$ has the solution $(w, 1, 2)$; hence $(u,v)_2 = 1$. The case $v \equiv 1 \pmod 4$ is symmetric.
Now suppose $u \equiv v \equiv -1 \pmod 4$. If $(u,v)_2 = 1$, let (z_0, x_0, y_0) be a primitive solution of $z^2 - ux^2 - vy^2 = 0$. Then

$$z_0^2 - ux_0^2 - vy_0^2 \equiv z_0^2 + x_0^2 + y_0^2 \pmod 4,$$

forcing $z_0 \equiv x_0 \equiv y_0 \equiv 0 \pmod 2$, which contradicts the assumption of primitivity. Hence, $(u,v)_2 = -1$.
Case $\alpha = 0, \beta = 1$. We must show $(2u, v)_2 = (-1)^{\epsilon(u)\epsilon(v) + \omega(v)}$. First, we show $(2,v)_2 = (-1)^{\omega(v)}$.
If $(2,v)_2 = 1$, let (z_0, x_0, y_0) solve $z^2 - 2x^2 - vy^2 = 0$ with $z_0, y_0 \not\equiv 0 \pmod 2$; then $z_0^2 \equiv y_0^2 \equiv 1 \pmod 8$. From

$$z_0^2 - 2x_0^2 - vy_0^2 \equiv 1 - 2x_0^2 - v \equiv 0 \pmod 8,$$

and $2x_0^2 \equiv 0$ or $2 \pmod 8$, we get $v \equiv \pm 1 \pmod 8$. If $v \equiv 1 \pmod 8$, then $(2,v)_2 = 1$ as expected. If $v \equiv -1 \pmod 8$, then $z^2 - 2x^2 + y^2 = 0$ has the solution $(1,1,1)$ modulo 8, and

(by Theorem 1 and Corollary 3 of Chapter 2.2, "Amelioration of approximate solutions," in Serre's *A Course in Arithmetic*) this lifts to a genuine solution, so $(2, v)_2 = 1$.
We next show $(2u, v)_2 = (u, v)_2(2, v)_2$. This holds automatically if $(2, v)_2 = 1$ or $(u, v)_2 = 1$, by Proposition 7.2. It remains to check the case $(2, v)_2 = (u, v)_2 = -1$, i.e., $v \equiv 3 \pmod 8$ and $u \equiv 3, 7 \pmod 8$.
If $v \equiv u \equiv 3 \pmod 8$, then

$$z^2 - 2ux^2 - vy^2 \equiv z^2 - 6x^2 - 3y^2 \equiv z^2 - 6x^2 + 5y^2 \pmod 8$$

has the solution $(1, 1, 1)$ modulo 8, so $(2u, v)_2 = 1$.
If $v \equiv 3 \pmod 8$ and $u \equiv 7 \pmod 8$, then

$$z^2 - 2ux^2 - vy^2 \equiv z^2 - 14x^2 - 3y^2 \equiv z^2 + 2x^2 - 3y^2 \pmod 8$$

has the solution $(1, 1, 1)$ modulo 8, so $(2u, v)_2 = 1$.
In every case,

$$(2u, v)_2 = (u, v)_2 \cdot (2, v)_2 = (-1)^{\epsilon(u)\epsilon(v)+\omega(v)}.$$

Case $\alpha = \beta = 1$. We must show

$$(2u, 2v)_2 = (-1)^{\epsilon(u)\epsilon(v)+\omega(u)+\omega(v)}.$$

Indeed,

$$(2u, 2v)_2 = (2u, -4uv)_2 = (2u, -uv)_2 = (-1)^{\epsilon(u)\epsilon(-uv)+\omega(-uv)}.$$

Using $\epsilon(uv) \equiv \epsilon(u) + \epsilon(v) \pmod 2$ and $\omega(uv) \equiv \omega(u) + \omega(v) \pmod 2$, we compute

$$\epsilon(u)\epsilon(-uv) + \omega(-uv) \equiv -\epsilon(u)^2\epsilon(v) - \omega(u) - \omega(v) \equiv \epsilon(u)\epsilon(v) + \omega(u) + \omega(v) \pmod 2,$$

so

$$(2u, 2v)_2 = (-1)^{\epsilon(u)\epsilon(v)+\omega(v)} = (-1)^{\epsilon(u)\epsilon(v)+\omega(u)+\omega(v)}.$$

□

Corollary. *For* $a, a', b \in Q_p^*$,

$$(aa', b)_p = (a, b)_p \cdot (a', b)_p.$$

Proof. For $p = \infty$, this is immediate.
Write $a = p^\alpha u$, $b = p^\beta v$, $a' = p^\gamma w$, so $aa' = p^{\alpha+\gamma}uw$.
For $p \neq 2$,

$$\begin{aligned}(a, b)_p \cdot (a', b)_p &= (-1)^{\alpha\beta\cdot\epsilon(p)}\left(\frac{u}{p}\right)^\beta\left(\frac{v}{p}\right)^\alpha \cdot (-1)^{\beta\gamma\cdot\epsilon(p)}\left(\frac{w}{p}\right)^\beta\left(\frac{v}{p}\right)^\gamma \\ &= (-1)^{\beta(\alpha+\gamma)\cdot\epsilon(p)}\left(\frac{uw}{p}\right)^\beta\left(\frac{v}{p}\right)^{\alpha+\gamma} \\ &= (aa', b)_p.\end{aligned}$$

For $p = 2$,

$$\begin{aligned}(a,b)_2 \cdot (a',b)_2 &= (-1)^{\epsilon(u)\epsilon(v)+\alpha\omega(v)+\beta\omega(u)} \cdot (-1)^{\epsilon(w)\epsilon(v)+\gamma\omega(v)+\beta\omega(w)} \\ &= (-1)^{\epsilon(v)(\epsilon(u)+\epsilon(w))+\beta(\omega(u)+\omega(w))+(\alpha+\gamma)\omega(v)} \\ &= (-1)^{\epsilon(v)\epsilon(uw)+\beta\omega(uw)+(\alpha+\gamma)\omega(v)} \\ &= (aa',b)_2.\end{aligned}$$

□

Theorem 9 (Hilbert Quadratic Reciprocity Law)**.** *Let $a, b \in Q^*$. For all but finitely many $v \in V$, where V is the set of all primes and ∞, we have $(a,b)_v = 1$, and*

$$\prod_{v \in V} (a,b)_v = 1.$$

Proof. Since a and b can be decomposed into products of -1, 2, and odd primes, and the Hilbert symbol is bilinear, it suffices to check 7 cases.
Case 1. $(-1,-1)_2 = (-1)^{\frac{-1-1}{2}\cdot\frac{-1-1}{2}} = -1 = (-1,-1)_\infty$, and $(-1,-1)_p = (-1)^0 = 1$ for $p \neq 2$. Hence,

$$\prod_{v \in V} (-1,-1)_v = 1.$$

Case 2. $(-1,2)_2 = (-1)^0 = 1 = (-1,2)_\infty$, and $(-1,2)_p = (-1)^0 = 1$ for $p \neq 2$. Hence,

$$\prod_{v \in V} (-1,2)_v = 1.$$

Case 3. $(-1,p)_2 = (-1)^{\frac{p-1}{2}} = (-1,p)_p$ and $(-1,p)_\infty = 1$; for any prime $q \neq 2, p$, $(-1,p)_q = (-1)^0 = 1$. Hence,

$$\prod_{v \in V} (-1,p)_v = 1.$$

Case 4. $(2,2)_2 = (-1)^0 = 1 = (2,2)_\infty$, and $(2,2)_p = (-1)^0 = 1$ for every prime p. Hence,

$$\prod_{v \in V} (2,2)_v = 1.$$

Case 5. $(2,p)_\infty = 1$ and $(2,p)_2 = (-1)^{\omega(p)} = \left(\frac{2}{p}\right) = (2,p)_p$; for any prime $q \neq 2, p$, $(2,p)_q = 1$. Hence,

$$\prod_{v \in V} (2,p)_v = 1.$$

Case 6. $(p,p)_\infty = 1$, $(p,p)_p = (-1)^{\frac{p-1}{2}}$, and $(p,p)_2 = (-1)^{\frac{p-1}{2}\cdot\frac{p-1}{2}} = (-1)^{\frac{p-1}{2}}$; for any prime $q \neq 2$, $(p,p)_q = (-1)^0 = 1$. Hence,

$$\prod_{v \in V} (p,p)_v = 1.$$

Case 7. $(p,q)_l = (-1)^0 = 1 = (p,q)_\infty$ for any prime $l \neq p, q, 2$. Note that $(p,q)_2 = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}$, $(p,q)_p = \left(\frac{q}{p}\right)$, and $(p,q)_q = \left(\frac{p}{q}\right)$. By the Quadratic Reciprocity Law,

$$(p,q)_2 \cdot (p,q)_p \cdot (p,q)_q = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}} \left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = \left((-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}\right)^2 = 1.$$

Hence,

$$\prod_{v\in V} (p,q)_v = 1.$$

Combining all seven cases completes the proof of the theorem. □